

Text as Data

Class 1: Text Representation

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Introductions

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- Your name and disciplinary background
- Experience with programming and text analysis
- A research question you might want to study with text

What would you like to get out of this course?

The course

- Representing text and making preprocessing choices
- Measuring concepts with dictionaries and classifiers
- Exploring patterns with clustering and topic models
- Working with embeddings and language models

We will combine explanations with Python exercises and research examples.

Course materials on GitHub

github.com/PeterTolochko/CEU_TAD_2026

Bookmark this page for the class.

Books for the course

Grimmer, Roberts & Stewart (2022)

Text as Data

Research design and measurement with text.

Jurafsky & Martin

Speech and Language Processing

NLP methods and language models. Free third-edition draft.

James et al. (2023)

An Introduction to Statistical Learning

Python edition. Model fitting and evaluation. Free PDF.

Further reading and practical references

Manning, Raghavan & Schütze (2008)

Introduction to Information Retrieval

Word counts, TF-IDF, and vector similarity. Free online.

Cosma Rohilla Shalizi

Advanced Data Analysis

from an Elementary Point of View

Statistical reasoning and model evaluation. Free draft, R examples.

Documentation for the practical sessions

- [spaCy 101](#): tokens and linguistic annotations
- [scikit-learn: text feature extraction](#): count matrices and TF-IDF

Today

By the end of class, you should be able to:

- ① Explain what a text measure tells us about a research question
- ② Build a document-term matrix for a small corpus
- ③ Interpret similarity between document vectors
- ④ Identify information that a representation leaves out

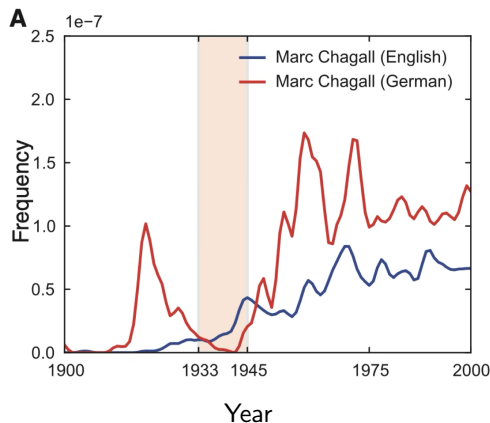
Research questions and text

Texts record what people choose to say or write.

- Speeches can reveal political attention
- News articles can reveal how an issue is framed
- Open survey responses can reveal concerns that fixed questions miss

Censorship in digitized books

Can changes in name frequency reveal censorship?



Texts

English and German books

Measure

Frequency of “Marc Chagall” by year

Pattern

German mentions nearly disappear during Nazi rule

What else could make a name disappear from books?

Michel et al. (2011), *Science*, Figure 4A.

Policy adoption and newspaper framing

Do policies adopted elsewhere predict how newspapers discuss smoking restrictions?

Texts: over 52,000 newspaper paragraphs
49 US states, 1996–2013

Result from the topic-model analysis

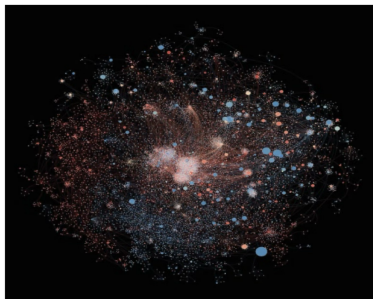
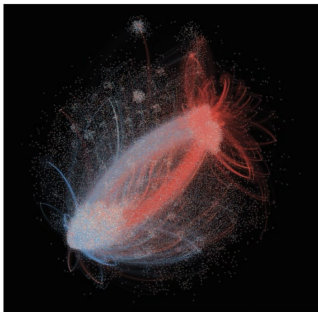
Framing of the policy	Earlier adoptions elsewhere
Practical consequences	Predictive relationship
Normative justifications	No predictive relationship

Gilardi, Shipan & Wüest (2021), *AJPS*.

Is online communication an echo chamber?

Barberá et al. (2015): does information travel across ideological boundaries?

Nearly 150 million tweets on 12 political and nonpolitical issues; estimated ideology for 3.8 million users.



Political issues showed more exchange among ideologically similar users; many nonpolitical events did not.

Barberá et al. (2015), *Tweeting From Left to Right*, Psychological Science.

Our example today

How much attention do US presidents give to jobs in their State of the Union speeches?

We will start by reading passages, then build and check a simple word-count indicator.

Jobs in presidential speeches

Barack Obama, 2010

"That is why jobs must be our number-one focus in 2010, and that's why I'm calling for a new jobs bill tonight."

Donald Trump, 2020

"Jobs are booming, incomes are soaring, poverty is plummeting, crime is falling, confidence is surging, and our country is thriving and highly respected again."

Do these passages express the same thing about jobs?

A concept and a first indicator

Concept: attention to employment in a speech

First indicator: occurrences of job or jobs

$$\text{mentions per 1,000 tokens} = \frac{\text{count of job + jobs}}{\text{number of tokens}} \times 1,000$$

- A longer speech has more opportunities to mention a word
- A rate makes speeches of different lengths easier to compare

Which discussions of employment would this count miss?

Checking the indicator against the text

Another use of “jobs” in Obama’s 2010 speech

“That’s why we’ve excluded lobbyists from policymaking jobs or seats on Federal boards and commissions.”

- Read passages that the indicator counts
- Look for relevant passages that it misses
- Decide what counts as attention to employment

Would you include this passage? Why?

The corpus and its metadata

A **corpus** is the collection of texts we analyse.

Our small sample contains four complete speeches:

Year	President	Party
2008	George W. Bush	Republican
2010	Barack Obama	Democratic
2016	Barack Obama	Democratic
2020	Donald Trump	Republican

One row per speech.

The unit of analysis

The same collection can produce different datasets.

One row per	Possible question
Speech	How much of each speech concerns employment?
Paragraph	Which policy issues appear alongside employment?
Sentence	Is this sentence positive or negative about jobs?

The unit determines which pieces of context stay together.

What can this corpus tell us?

- These texts record a particular public communication setting
- Four selected speeches cannot establish a historical trend
- Differences may reflect the speaker, the year, or the situation
- The sample does not isolate a party effect

What additional texts would you want for the jobs question?

What is Text?

- Data
- Unstructured
- (Highly) multidimensional
- Difficult to work with (if you're not human)

From Text to Structure

- We need to structure the text before analysis
- Different ways to represent text so computers “understand”
- Different ways to model text so that both we and the computer “understand”
- Different research questions → different representations

A tiny corpus

- ① John loves ice-cream
- ② John loves oranges
- ③ Mary hates ice-cream

For this example, we:

- use lowercase words
- keep ice-cream as one token

How many word occurrences? How many distinct words?

Tokens, types, and vocabulary

Tokens: individual occurrences in the text

john loves ice-cream john loves oranges
mary hates ice-cream

Types: distinct tokens in the corpus

john, loves, ice-cream, oranges, mary, hates

9 tokens. 6 types. These six types form our vocabulary.

Exercise: a document-term matrix

Rows are documents. Columns are terms. Cells contain counts.

	john	loves	ice-cream	oranges	mary	hates
D_1	?	?	?	?	?	?
D_2	?	?	?	?	?	?
D_3	?	?	?	?	?	?

- ① John loves ice-cream
- ② John loves oranges
- ③ Mary hates ice-cream

Fill in the matrix with a neighbour. You have two minutes.

The completed matrix

	john	loves	ice-cream	oranges	mary	hates
D_1	1	1	1	0	0	0
D_2	1	1	0	1	0	0
D_3	0	0	1	0	1	1

- Row sums give document length in tokens
- Column sums give each term's frequency in the corpus
- A zero means that a term does not occur in that document

Dimensions and sparsity

$$X \in \mathbb{R}^{N \times K}, \quad X_{dv} = \text{count of term } v \text{ in document } d$$

- N is the number of documents
- K is the vocabulary size
- Our matrix has $3 \times 6 = 18$ cells, nine of them zero

In larger corpora, a document usually contains only a small share of the vocabulary. Many cells are zero.

We call a matrix with mostly zero entries **sparse**.

Bag-of-words and word order

A: Police confront protesters

B: Protesters confront police

	police	confront	protesters
A	1	1	1
B	1	1	1

Bag-of-words keeps counts and discards word order.

Can this matrix tell us who confronts whom?

What bag-of-words preserves

Available in the matrix

- Which words occur
- How often they occur
- Vocabulary shared by documents

Missing from the matrix

- Word order
- Who did what to whom
- The scope of negation

Word counts may help us measure attention to an issue. Some questions require more context.

Vector Space Model

Represent texts as vectors in a multidimensional space.

- A **dimension** is one feature used to describe an object
- A **coordinate** is the object's value on that feature
- A **vector** lists these coordinates in a fixed order

Start with familiar spaces, then replace physical coordinates with word counts.

Multidimensional Spaces

How many coordinates do we need to describe a location or event?

Dimensions	Coordinates
2D: on a map	Longitude, latitude
3D: in space	Longitude, latitude, height
4D: an event	Longitude, latitude, height, time

Adding a dimension adds another value to the description.

From geographic coordinates to word counts

For a location, coordinates describe **where it is**.

For a document, coordinates describe **which words it contains**.

With the vocabulary ordered as

(john, loves, ice-cream, oranges, mary, hates),

“John loves ice-cream” becomes

$$x = (1, 1, 1, 0, 0, 0).$$

- Six terms $\rightarrow$ six dimensions
- K terms $\rightarrow K$ coordinates for every document
- More documents add points; new terms add dimensions

Documents as vectors

Each row of the matrix is a document vector.

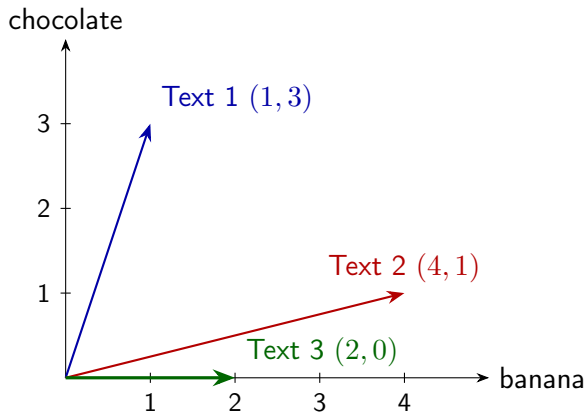
For a vocabulary of {banana, chocolate}:

	banana	chocolate
Text 1	1	3
Text 2	4	1
Text 3	2	0

Text 1 is “chocolate chocolate chocolate banana”.

Its coordinates are (1, 3).

Three documents in two dimensions



Which pair has the most similar balance of words?

Document length and word proportions

Suppose we repeat every word in Text 2.

$$x = (4, 1), \quad 2x = (8, 2)$$

	Original	Repeated
Tokens	5	10
Banana share	0.80	0.80
Chocolate share	0.20	0.20

The vector becomes longer. Its direction stays the same.

Cosine similarity

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$$

- Compares the directions of two nonzero vectors
- Equals 1 when they point in the same direction
- Equals 0 for count vectors with no shared terms

For nonnegative word counts, cosine ranges from 0 to 1.

$$\cos(x, 2x) = 1$$

A worked comparison

Text 2 is (4, 1). Text 3 is (2, 0).

$$\cos(T_2, T_3) = \frac{4 \times 2 + 1 \times 0}{\sqrt{4^2 + 1^2} \sqrt{2^2 + 0^2}} = \frac{8}{2\sqrt{17}} \approx 0.970$$

Pair	Cosine similarity
Text 1 and Text 2	0.537
Text 1 and Text 3	0.316
Text 2 and Text 3	0.970

This measures similarity in word counts. Its meaning depends on the features we chose.

The matrix in Python

```
from collections import Counter

texts = ["john loves ice-cream",
         "john loves oranges",
         "mary hates ice-cream"]

vocab = ["john", "loves", "ice-cream",
         "oranges", "mary", "hates"]
counts = [Counter(text.split()) for text in texts]
X = [[row[word] for word in vocab] for row in counts]
```

`split()` separates these toy texts at spaces.
`Counter` records how often each token occurs.

Practical: counts and meaning

Open `meeting_01.ipynb`, Parts 1 and 2.

- ① Run the John/Mary example. Check it against your matrix.
- ② Change the first sentence to “john loves loves ice-cream”. Predict which cell and row total will change.
- ③ Run the police/protesters example. Explain why its rows match.

Which question could you answer using these counts?

Practical: a different representation

In Part 3, compare the banana/chocolate documents.

- ① Predict what happens when you repeat all of Text 2.
- ② Calculate cosine similarity using word counts.
- ③ Switch to **binary presence**: 1 if a word occurs, 0 otherwise.

Binary vectors for Texts 1 and 2 are both $(1, 1)$.

What information did this change remove?

The indicator in four real speeches

	Tokens	job + jobs	Per 1,000
Bush 2008	5,678	6	1.06
Obama 2010	7,190	29	4.03
Obama 2016	6,030	19	3.15
Trump 2020	6,232	14	2.25

Part 4 of the notebook reproduces this table and shows the counted passages.

What would you check before calling this “attention to employment”?

Discussion

- ① What changed when you replaced counts with presence?
- ② Which information disappeared before you calculated similarity?
- ③ Which representation would help answer your own research question?

A calculation can be correct while the measure is poorly suited to the question.

Later: embeddings and language models

Our count vectors have one dimension for each vocabulary item.

Later, we will work with learned representations:

- Embeddings can place words or texts with different vocabulary near each other
- Contextual representations use surrounding words

We will still need to decide what to measure and how to check it.

Next class: preprocessing

How do tokenization, case, stop words, and phrases change our features?

An example to think about

"I support the policy."

"I do not support the policy."

What happens if we discard I, do, not, and the?